

Is there a Hamiltonian inside a decision model?

Probing the energy landscape of Jev for quantum chemistry

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Abstract

Jev is a proprietary “System One” model that returns calibrated probability distributions over a fixed, typed answer schema instead of text, and answers many questions about one input in a single parallel pass. Any such output defines local energies $E = -\log p$, so it is natural to ask whether these energies form one Hamiltonian-like landscape and whether that landscape is useful for quantum chemistry. We test this through black-box access to `jev-1.13.0` with 2,235 distinct queries, using PySCF for ground truth. The conditionals are close to, but not exactly, those of a single joint distribution: the logically forced coupling is symmetric, yet 57% of inferred pairwise couplings are asymmetric beyond three bootstrap standard errors, and complementary statements do not sum to one. Given a Hamiltonian written in the prompt, Jev does not track its spectrum. Its preferences over Ising configurations follow the signs of the printed coefficients, so rewriting the same energies in the textbook sign convention reverses the rank correlation (from -0.41 to $+0.34$ for four spins), and its energy read-out for H_2 is off by $104 \text{ m}E_h$ on average. On quantum-chemistry decisions it ranks multireference character, T_1 diagnostics and RHF instability with AUROC 0.74–0.89, comparable to a two-feature logistic model, but it is poorly calibrated and falls to 0.60–0.68 when the molecule’s name is removed. As an importance prior for selected configuration interaction it beats random selection in all ten systems and loses to a first-order perturbative estimate in nine. Jev therefore behaves as a fast triage layer driven by textual cues, and its outputs should not be read as a physical energy landscape.

1 Introduction

A classifier that outputs a probability distribution $p(y \mid x)$ can always be rewritten as a Gibbs distribution $p(y \mid x) \propto \exp[-E(x, y)]$ with $E = -\log p$ up to a constant [1, 2]. For one softmax head the rewriting says nothing new. It says more when a model answers *several* questions about the same input, because the per-question energies may or may not assemble into one landscape over joint answers, as the conditionals of a Boltzmann machine do [3]. Only a single landscape can be sampled, minimised or added to a physical Hamiltonian, the uses that Ising and QUBO encodings make routine [4, 5].

Jev [6, 7] makes this question concrete. It accepts text or structured state and a set of typed questions (a *yes/no Noul*, a *Choice* over at most 255 options, or a *Score* over 2–10 ordered levels) and returns a probability for every allowed answer, trained, according to its developer, for calibration against outcomes rather than for human preference. Questions are evaluated in parallel, latency is well under a second, and outputs cannot fall outside the schema [8]. A quantum-chemistry workflow makes many small typed decisions of this kind before any expensive calculation: is the system multireference, which spin state is lowest, which determinants matter. The developer also warns that structural invariants “aren’t guaranteed” and that the model “is not a calculator” [9].

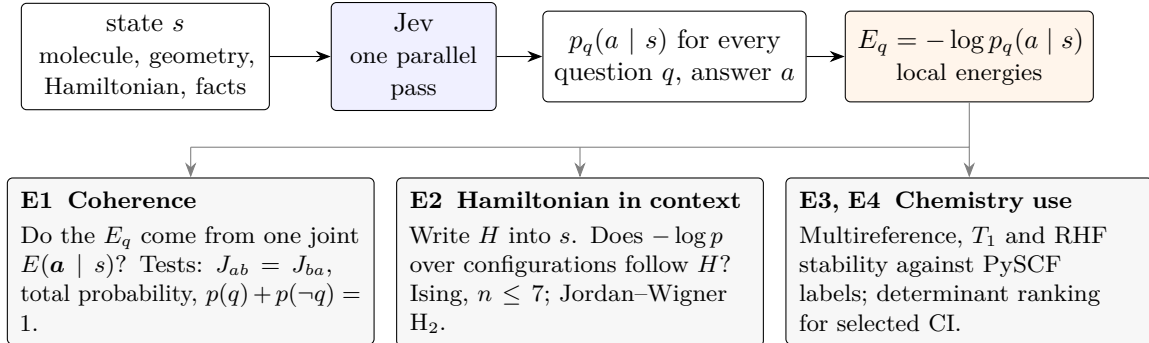


Figure 1: Study design. Jev maps a state and a set of typed questions to answer distributions in one pass. Reading each distribution as a local energy, we test whether the local energies are consistent with one landscape (E1), whether they reproduce a Hamiltonian supplied in the input (E2), and whether they are useful for quantum-chemistry decisions (E3, E4).

We ask three questions and answer each with measurements against physical ground truth. (i) Do Jev’s conditionals admit a single joint landscape? We use the classical compatibility conditions for conditional distributions [10, 11] to test coupling symmetry and total probability. (ii) Can Jev represent a Hamiltonian given in context? We give it random classical Ising models and the Jordan–Wigner qubit Hamiltonian of H_2 [12, 13] and compare its preferences with the exact spectra. (iii) Is it useful for real quantum-chemistry decisions? We score its predictions of multireference diagnostics against PySCF labels with proper scoring rules [14], and use it to rank determinants for selected configuration interaction [15, 16]. Figure 1 summarises the design.

Our contributions are (1) a black-box protocol, based on coupling symmetry, total probability and negation, for testing whether a multi-query decision model has one energy landscape; (2) evidence that Jev’s landscape is approximately but measurably incoherent, and that its view of an in-context Hamiltonian follows printed coefficient signs rather than energies; and (3) a quantitative account of where it helps and where it does not in a quantum-chemistry pipeline. All code, cached API responses and figures are released.

2 Background and related work

Classifiers as energy models. Energy-based models assign a scalar energy to each configuration and convert it to probability through a Gibbs distribution [1]. Hopfield networks and Boltzmann machines give such energies pairwise, symmetric couplings [3, 17], and attention can be read as the update of a continuous Hopfield energy [18]. Grathwohl et al. [2] showed that the logits of any softmax classifier define a joint energy over inputs and labels, and Winer and Hanin [19] studied a random network as a Hamiltonian over its inputs. These works treat one output head; Jev exposes many typed heads over one shared state, which is the setting where a joint landscape can fail to exist.

When do conditionals have a joint? Local conditional specifications define a joint distribution only under compatibility conditions [10, 11, 20]. Separately learned conditionals, as in dependency networks, are generally incompatible [21]. For binary variables compatibility has a simple necessary consequence: the log odds ratio between two variables is symmetric, so the coupling inferred from b ’s effect on a must equal that inferred from a ’s effect on b . Recent work finds logical incoherence in the probabilities of language models and forecasters [22–24] and proposes repairs [25, 26]; these

studies test identities on generative models, not the joint structure of a parallel, native-distribution model.

Calibration. Deep networks are often miscalibrated [27]; we report expected calibration error [28] and the Brier score [29], a strictly proper rule [14]. Large language models are reasonably calibrated on some formats [30, 31]. Calibration of each marginal does not imply joint coherence, which is why we test both.

Hamiltonians and quantum chemistry. Molecular Hamiltonians are mapped to qubits by the Jordan–Wigner transformation [12] and solved variationally [13, 32], on annealers [33, 34], or with neural wavefunctions [35–37]; see [38, 39] for reviews. Learning a Hamiltonian from data assumes the data come from a Gibbs state [40–42]. Before a calculation, practitioners decide whether a system is multireference, using diagnostics such as T_1 [43, 44], choose active spaces [45, 46], and select determinants [15, 16, 47]. Machine-learned models for these decisions are task-specific and trained on computed descriptors [48–50]. General language models are used in chemistry as tool-using agents [51, 52], where they are knowledgeable but overconfident [53] and fragile on numerical reasoning [54].

3 Setup

Access and energies. We queried `jev-1.13.0` through the public API on 2026-09-23 and cached every request and response. Probabilities are returned to two decimals and exact zeros occur, so we clip to $[0.005, 0.995]$ before taking logarithms and define $E_q(a \mid s) = -\log p_q(a \mid s)$ and log odds $\ell = \text{logit } p$. Identical requests do not return identical probabilities: across repeats the standard deviation of a Noul is 0.013 on average (up to about 0.03). In a pilot on one molecule, answering a question alone or alongside three others changed its probability by no more than this noise. We therefore repeat every query three times and average. Median latency was 0.58 s (90th percentile 0.82 s), and all 2,235 unique requests together cost US\$0.09.

Chemistry ground truth. Twelve molecules (H_2 , LiH , HF , N_2 , F_2 , CO , BH , BeO , C_2 , H_2O , NH_3 , CH_4) were each placed at 1.0, 1.5, 2.0 and 2.5 times their equilibrium bond lengths, giving 48 cases. With PySCF in the cc-pVDZ basis we computed RHF, a stability-analysed UHF, CASCI on RHF orbitals (six orbitals, up to six active electrons) and CCSD. Three binary labels follow: *multireference* if the largest CASCI coefficient squared $c_0^2 < 0.9$ (positive rate 0.58), T_1 if the CCSD T_1 diagnostic exceeds 0.02 [43] (0.62), and *RHF unstable* if UHF lies more than 0.1 mE_h below RHF (0.77). Each prompt gives the molecule name, Cartesian geometry, basis, and a note stating the stretch factor.

Pre-registration. The analyses of E1–E4 were written down after a ten-call pilot and before the main data were collected; checks added after seeing data are labelled exploratory. The plan and its amendments are in the repository.

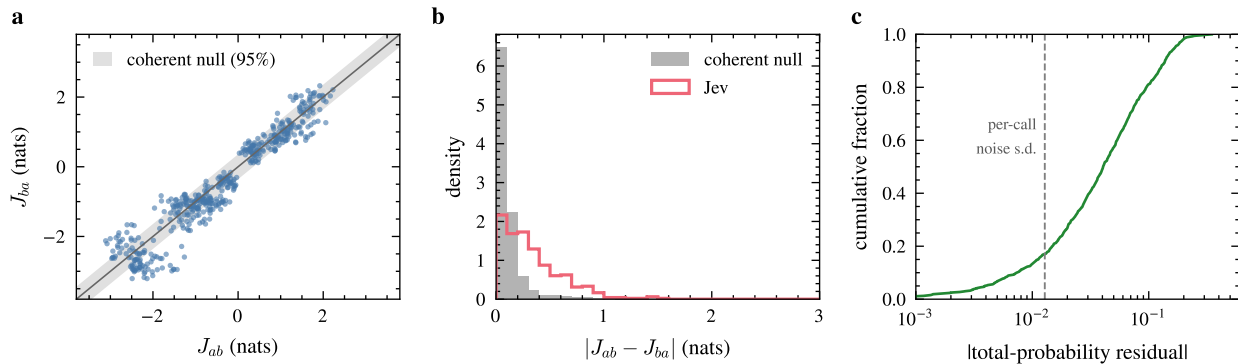


Figure 2: Coherence of Jev’s conditionals (E1; 48 molecular cases, 10 statement pairs, 3 repeats). **a**, Coupling inferred in each direction; a joint distribution requires points on the diagonal. The band is the 95% range of a coherent model with Jev’s rounding and noise. **b**, Distribution of the asymmetry $|J_{ab} - J_{ba}|$ for Jev and for the coherent null. **c**, Cumulative distribution of violations of the law of total probability; the dashed line is the mean per-call noise.

4 Results

4.1 E1: the conditionals are close to one landscape, but not exactly

For five statements about each case (multireference; RHF qualitatively correct; $T_1 > 0.02$; RHF unstable; CCSD(T) within 1 kcal/mol of full CI) we measured each marginal $p(a)$ and each conditional $p(a | b = v)$, stating b as an “established fact” in the input. If the answers came from one joint distribution over the five binary statements, the pairwise coupling

$$J_{ab} = \ell(p(a | b=1)) - \ell(p(a | b=0)) \quad (1)$$

would equal J_{ba} , because both equal the log odds ratio of the pair, and the law of total probability $p(a) = \sum_v p(a | b=v) p(b=v)$ would hold.

The couplings are large and mostly have the right structure (Fig. 2a). The median $|J|$ is 1.10 nats, and the logically forced pair “multireference” and “RHF adequate” is strongly and symmetrically anticorrelated ($J = -2.43$ and -2.35). Symmetry nevertheless fails for many pairs. The median asymmetry $|J_{ab} - J_{ba}|$ over the 480 pair-case combinations is 0.26 nats, and 57% exceed three bootstrap standard errors, our pre-registered criterion. Because a bootstrap over three repeats can understate the noise, we also simulated a perfectly coherent two-variable Ising model passed through Jev’s rounding and noise; 35% of Jev’s pairs lie beyond the 95th percentile of that null (Fig. 2b). Total probability is violated by a median of 0.041 and by more than 0.138 in a tenth of cases, well above the per-call noise (Fig. 2c). In an exploratory check, the probabilities of a statement and its explicit negation, asked together, summed to between 0.89 and 1.25; for two of the five statements more than two thirds of cases missed 1 by over 0.1. Jev’s outputs behave like a dependency network [21]: a set of approximately compatible local energies with no exact joint behind them.

4.2 E2: an in-context Hamiltonian is read by the signs of its coefficients

We gave Jev random classical Ising Hamiltonians $H(\mathbf{s}) = \sum_i h_i s_i + \sum_{i<j} J_{ij} s_i s_j$ with $h_i, J_{ij} \sim U[-1, 1]$ on $n = 3-7$ spins (ten instances per n) and asked, as a Choice over all 2^n configurations, which one minimises H . A model with a usable landscape should put more probability on lower-energy configurations. Jev did the opposite: the Spearman correlation between $p(\mathbf{s})$ and $-H(\mathbf{s})$

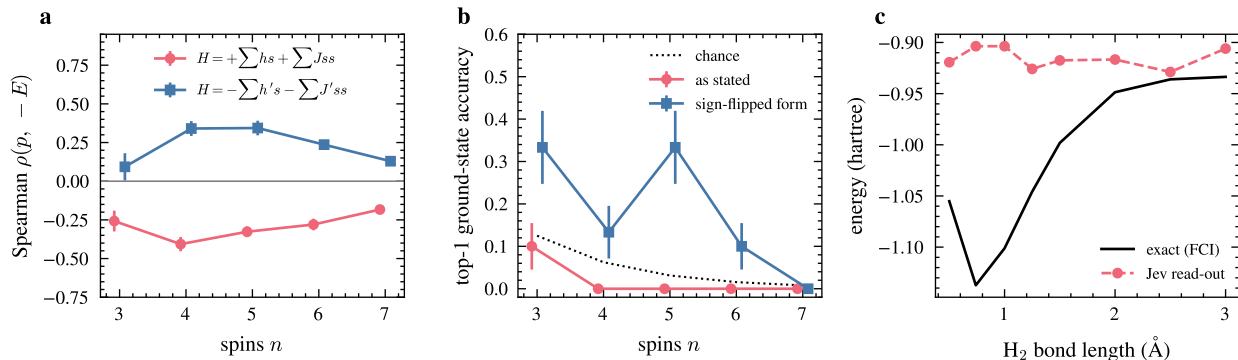


Figure 3: Jev given a Hamiltonian in its input (E2). **a**, Rank correlation between Jev’s probability for each spin configuration and its negative energy, for random Ising models written as stated (circles) and for the same energies written with flipped coefficient signs (squares); mean $\pm$ s.e.m. over 10 instances $\times$ 3 repeats. **b**, Top-1 ground-state accuracy; dotted line, chance 2^{-n} . **c**, Exact ground-state energy of the Jordan–Wigner H_2 /STO-3G Hamiltonian and Jev’s read-out from a ten-level Score.

ranged from -0.18 to -0.41 (Fig. 3a), and top-1 accuracy was at or below chance for every n (Fig. 3b). Decomposing the energy shows why: p correlated with $+\sum_i h_i s_i$ ($\rho = 0.29$ – 0.51) and hardly at all with the coupling term ($|\rho| \leq 0.21$). Jev aligns each spin with the sign of its printed field, which maximises the field energy.

This suggests that Jev reads the sign of each coefficient and ignores the functional form, perhaps because physics convention usually writes $H = -\sum h s - \sum J s s$. To test this we presented exactly the same energies in that convention, with $h' = -h$ and $J' = -J$ (exploratory). The correlation reversed to between $+0.09$ and $+0.34$, and top-1 accuracy rose above chance for $n \leq 6$ (Fig. 3a,b). A relabelling that leaves every energy unchanged flips Jev’s preferences, so its landscape depends on how the Hamiltonian is written and not only on the Hamiltonian.

The Jordan–Wigner Hamiltonian of H_2 in STO-3G gave a consistent picture. Asked which of the 16 basis states carries the largest ground-state weight, Jev chose the Hartree–Fock state $|1100\rangle$ in 16 of 24 queries and otherwise chose $|0011\rangle$, which has the second-largest weight. The probability it gave $|1100\rangle$ stayed near 0.2 at every bond length, whereas the true weight falls from 0.995 at 0.5 Å to 0.54 at 3.0 Å. Its ground-state energy, read from a ten-level Score spanning -1.20 to -0.75 E_h , varied by only 25 mE_h across bond lengths where the exact energy varies by 204 mE_h , with a mean absolute error of 104 mE_h (Fig. 3c). Chemical accuracy is 1.6 mE_h .

4.3 E3: useful ranking of multireference character, poor calibration

Table 1 scores Jev’s marginal probabilities on the 48 cases. Its ranking is informative: AUROC is 0.74 for multireference character, 0.85 for $T_1 > 0.02$ and 0.89 for RHF instability. A leave-one-out logistic regression on two numbers available after an RHF calculation, the stretch factor and the HOMO–LUMO gap, reached 0.85, 0.83 and 0.92. Its probabilities are less accurate: the Brier score is worse than the two-feature model’s for all three labels, the expected calibration error ranges from 0.13 to 0.28 (Fig. 4a), and for RHF instability the Brier score (0.184) is worse even than that of the constant base rate (0.177).

To find what drives the ranking we removed the molecule name and the stretch note and gave only Cartesian coordinates (exploratory). AUROC fell to 0.60–0.68 and the Brier score rose to 0.31–0.35, worse than the base rate for every label. Most of Jev’s skill comes from recognising a named molecule and a stated stretch, the same textual prior a chemist would use; it reads little

Table 1: Quantum-chemistry decisions on 48 cases (E3). Higher AUROC and lower Brier score and ECE are better. Logistic models are fitted leave-one-out. "Geometry only" removes the molecule name and stretch note from Jev’s input.

Predictor	Multireference ($c_0^2 < 0.9$)			$T_1 > 0.02$			RHF unstable		
	AUROC	Brier	ECE	AUROC	Brier	ECE	AUROC	Brier	ECE
Base rate	0.50	0.243	–	0.50	0.234	–	0.50	0.177	–
Logistic (stretch)	0.79	0.156	0.12	0.76	0.166	0.24	0.87	0.108	0.19
Logistic (stretch, gap)	0.85	0.149	0.14	0.83	0.154	0.19	0.92	0.103	0.14
Jev	0.74	0.207	0.13	0.85	0.168	0.16	0.89	0.184	0.28
Jev, geometry only	0.60	0.311	–	0.64	0.308	–	0.68	0.346	–

from the coordinates.

4.4 E4: a weak but real prior for selected configuration interaction

Selected CI keeps only the determinants that matter [15, 16]. We tested whether Jev can rank them. For ten systems we built the 100-determinant CAS(6e,5o) space on RHF orbitals, gave Jev the active orbital energies and the list of determinants in occupation notation, and asked which non-reference determinant has the largest weight in the exact ground state. Its Choice probabilities define a ranking; we added determinants in that order to the reference, diagonalised the Hamiltonian in each subspace with a spin penalty that keeps the lowest root a singlet, and recorded the error against full CASCI. We compared this with the oracle ordering by exact $|c_i|^2$, a first-order perturbative estimate $|\langle D_i | H | D_0 \rangle|^2 / (H_{ii} - H_{00})$, ordering by excitation rank and orbital-energy cost, ordering by diagonal energy H_{ii} , and random ordering.

Jev’s probabilities correlate positively with the exact weights in all ten systems (Spearman $\rho = 0.15$ – 0.49). Averaged over the whole curve, Jev’s ordering reduces the error below random ordering in 10 of 10 systems (paired Wilcoxon $p = 0.002$) and below diagonal ordering in 9 of 10 ($p = 0.004$), but it is not distinguishable from the excitation-rank heuristic (7 of 10, $p = 0.08$) and is worse than the first-order estimate in 9 of 10 ($p = 0.014$; Fig. 4b,c). The first-order estimate is computed from matrix elements that any selected-CI code already has. Part of Jev’s disadvantage is resolution: it returns exactly zero for about 60% of the 99 options, leaving only 14–18 distinct probability values, so most of the ranking is decided by tie-breaking, which we average over 30 random orders.

5 Discussion

Jev’s outputs define local energies, and within one domain these energies are close to mutually consistent: large couplings have the expected sign and the strongest logical coupling is symmetric. The remaining incoherence is small in any one query but systematic, which is what one expects when each head is trained to be calibrated on its own [21]; calibrated marginals need not share a joint. A Gibbs sampler run on such conditionals converges to a distribution that depends on the order of updates, so there is no single energy whose minima one could search for.

Given a Hamiltonian as text, Jev does not compute with it and instead responds to surface features, here the sign of each coefficient. The convention-flip experiment rules out the simpler explanation that Jev knows the energies but orders them wrongly, because relabelling that preserves every energy reversed its preferences. This agrees with the developer’s statement that Jev is not a

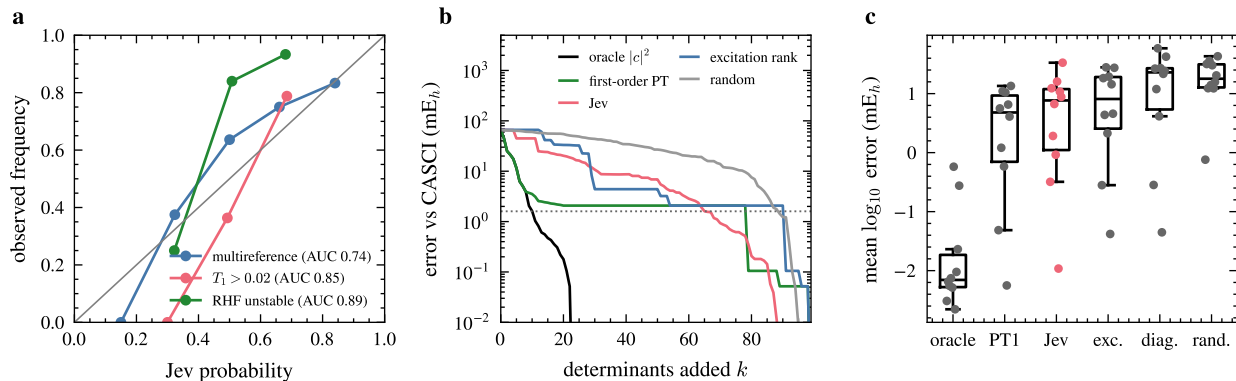


Figure 4: Jev in quantum-chemistry workflows. **a**, Reliability diagram for the three E3 labels (five equal-width bins; empty bins omitted). **b**, Selected-CI error against the number of added determinants for N_2 at equilibrium, CAS(6e,5o)/cc-pVDZ; the dotted line is chemical accuracy (1.6 mE_h). Jev and random curves are geometric means over 30 tie-break or permutation draws. **c**, Mean log₁₀ error over the whole curve for ten systems; lower is better.

calculator [9] and with the fragility of language models on numerical reasoning [54]. A model that cannot rank the eight configurations of a three-spin Ising model cannot stand in for an annealer or a variational solver [4, 32], and its energies are two orders of magnitude from chemical accuracy.

Jev is more useful for triage. It ranks multireference character and RHF instability about as well as a small descriptor model and returns in well under a second, at a cost measured in thousandths of a cent. It could route molecules between single- and multireference methods before any calculation, provided its probabilities are recalibrated on local data and a human or a cheap diagnostic checks the borderline cases. The geometry-only experiment limits this use to named, textbook systems; the skill should not be expected to carry over to unfamiliar molecules described only by coordinates. For determinant selection it is dominated by a free perturbative estimate and should not be used.

Limitations. We studied one model version through a black box, and later versions may behave differently. Prompt wording may account for some failures; the convention-flip and geometry-only controls separate the two in those experiments, but the other prompts were not varied. The chemistry set is small (48 cases, twelve closed-shell molecules, one basis), its labels depend on thresholds, and the selected-CI test uses 100-determinant spaces that are far from production scale. Coherence was tested on pairs; higher-order and cycle conditions were not tested, so our incoherence estimates are lower bounds. The two-decimal rounding and the sampling noise of the API limit the resolution of every energy we report.

6 Conclusion

Jev’s typed probabilities can be read as energies, but not as one Hamiltonian. The local energies are approximately compatible, they do not follow a physical Hamiltonian supplied in the input, and they carry a textual prior about molecules that is useful for fast triage and weak for determinant selection. The compatibility tests used here need only black-box access, and they can be applied to any model that answers many questions about one input.

Code and data. Code, the pre-registered plan, all cached API responses and the PySCF ground truth are at <https://github.com/dongzhaoh321418-lab/jev-hamiltonian>. Every analysis and

figure can be regenerated offline from the cache.

Use of AI tools. The experiments, analysis code and a first draft of the text were produced with the assistance of an AI coding assistant (Anthropic’s Claude). The author directed the study, checked the results against the released data and takes responsibility for the content.

References

- [1] Yann LeCun, Sumit Chopra, Raia Hadsell, Marc’Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In Gökhan Bakır, Thomas Hofmann, Bernhard Schölkopf, Alexander J. Smola, and Ben Taskar, editors, *Predicting Structured Data*. MIT Press, 2006. URL <http://yann.lecun.com/exdb/publis/pdf/lecun-06.pdf>.
- [2] Will Grathwohl, Kuan-Chieh Wang, Jörn-Henrik Jacobsen, David Duvenaud, Mohammad Norouzi, and Kevin Swersky. Your classifier is secretly an energy based model and you should treat it like one, 2019. URL <https://arxiv.org/abs/1912.03263>.
- [3] David H. Ackley, Geoffrey E. Hinton, and Terrence J. Sejnowski. A learning algorithm for Boltzmann machines. *Cognitive Science*, 9(1):147–169, 1985. doi: 10.1207/s15516709cog0901_7.
- [4] Andrew Lucas. Ising formulations of many NP problems. *Frontiers in Physics*, 2:5, 2014. doi: 10.3389/fphy.2014.00005.
- [5] Tadashi Kadowaki and Hidetoshi Nishimori. Quantum annealing in the transverse Ising model. *Physical Review E*, 58(5):5355–5363, 1998. doi: 10.1103/PhysRevE.58.5355.
- [6] Diogo Almeida. Introducing system one models & Jev. TypeSafe AI Blog, September 2026. URL <https://typesafe.ai/blog/introducing-system-one-models-and-jev>. Published 2026-09-15.
- [7] TypeSafe AI. API reference. TypeSafe AI documentation, 2026. URL <https://docs.typesafe.ai/api>.
- [8] Simon Willison. Jev introduces a new shape of LLM—system one, aka decision models. Blog post, simonwillison.net, September 2026. URL <https://simonwillison.net/2026/Sep/21/jev/>. Published 2026-09-21.
- [9] TypeSafe AI. Jev 1.13 jaggedness. TypeSafe AI documentation, 2026. URL <https://docs.typesafe.ai/model-jaggedness/jev-1.13>.
- [10] Julian Besag. Spatial interaction and the statistical analysis of lattice systems. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):192–225, 1974. doi: 10.1111/j.2517-6161.1974.tb00999.x.
- [11] Barry C. Arnold and S. James Press. Compatible conditional distributions. *Journal of the American Statistical Association*, 84(405):152–156, 1989. doi: 10.1080/01621459.1989.10478750.
- [12] P. Jordan and E. Wigner. Über das Paulische äquivalenzverbot. *Zeitschrift für Physik*, 47(9–10):631–651, 1928. doi: 10.1007/BF01331938.

- [13] P. J. J. O’Malley, R. Babbush, I. D. Kivlichan, J. Romero, J. R. McClean, R. Barends, J. Kelly, P. Roushan, A. Tranter, N. Ding, B. Campbell, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, A. G. Fowler, E. Jeffrey, E. Lucero, A. Megrant, J. Y. Mutus, M. Neeley, C. Neill, C. Quintana, D. Sank, A. Vainsencher, J. Wenner, T. C. White, P. V. Coveney, P. J. Love, H. Neven, A. Aspuru-Guzik, and J. M. Martinis. Scalable quantum simulation of molecular energies. *Physical Review X*, 6(3):031007, 2016. doi: 10.1103/PhysRevX.6.031007.
- [14] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [15] B. Huron, J. P. Malrieu, and P. Rancurel. Iterative perturbation calculations of ground and excited state energies from multiconfigurational zeroth-order wavefunctions. *The Journal of Chemical Physics*, 58(12):5745–5759, 1973. doi: 10.1063/1.1679199.
- [16] Adam A. Holmes, Norm M. Tubman, and C. J. Umrigar. Heat-bath configuration interaction: An efficient selected configuration interaction algorithm inspired by heat-bath sampling. *Journal of Chemical Theory and Computation*, 12(8):3674–3680, 2016. doi: 10.1021/acs.jctc.6b00407.
- [17] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.
- [18] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need, 2020. URL <https://arxiv.org/abs/2008.02217>.
- [19] Mike Winer and Boris Hanin. Deep neural nets as Hamiltonians, 2025. URL <https://arxiv.org/abs/2503.23982>.
- [20] J. M. Hammersley and P. Clifford. Markov fields on finite graphs and lattices. Unpublished manuscript, 1971. URL <https://ora.ox.ac.uk/objects/uuid:4ea849da-1511-4578-bb88-6a8d02f457a6>.
- [21] David Heckerman, David Maxwell Chickering, Christopher Meek, Robert Rounthwaite, and Carl Kadie. Dependency networks for inference, collaborative filtering, and data visualization. *Journal of Machine Learning Research*, 1:49–75, 2000. URL <https://www.jmlr.org/papers/v1/heckerman00a.html>.
- [22] Lukas Fluri, Daniel Paleka, and Florian Tramèr. Evaluating superhuman models with consistency checks, 2023. URL <https://arxiv.org/abs/2306.09983>.
- [23] Daniel Paleka, Abhimanyu Pallavi Sudhir, Alejandro Alvarez, Vineeth Bhat, Adam Shen, Evan Wang, and Florian Tramèr. Consistency checks for language model forecasters. In *International Conference on Learning Representations (ICLR)*, 2025. URL <https://openreview.net/forum?id=r5IXBlTCGc>.
- [24] Jian-Qiao Zhu and Thomas L. Griffiths. Incoherent probability judgments in large language models, 2024. URL <https://arxiv.org/abs/2401.16646>. CogSci 2024.

- [25] Nora Kassner, Oyvind Tafjord, Hinrich Schütze, and Peter Clark. BeliefBank: Adding memory to a pre-trained language model for a systematic notion of belief. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 8849–8861, 2021. doi: 10.18653/v1/2021.emnlp-main.697.
- [26] Diego Calanzone, Stefano Teso, and Antonio Vergari. Towards logically consistent language models via probabilistic reasoning, 2024. URL <https://arxiv.org/abs/2404.12843>. ICLR 2024 Workshop on Reliable and Responsible Foundation Models.
- [27] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [28] Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, 2015. doi: 10.1609/aaai.v29i1.9602.
- [29] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [30] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know, 2022. URL <https://arxiv.org/abs/2207.05221>.
- [31] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5433–5442, 2023. doi: 10.18653/v1/2023.emnlp-main.330.
- [32] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5:4213, 2014. doi: 10.1038/ncomms5213.
- [33] Rongxin Xia, Teng Bian, and Sabre Kais. Electronic structure calculations and the Ising Hamiltonian. *The Journal of Physical Chemistry B*, 122(13):3384–3395, 2018. doi: 10.1021/acs.jpcc.7b10371.
- [34] Scott N. Genin, Ilya G. Ryabinkin, and Artur F. Izmaylov. Quantum chemistry on quantum annealers, 2019. URL <https://arxiv.org/abs/1901.04715>.
- [35] Giuseppe Carleo and Matthias Troyer. Solving the quantum many-body problem with artificial neural networks. *Science*, 355(6325):602–606, 2017. doi: 10.1126/science.aag2302.
- [36] David Pfau, James S. Spencer, Alexander G. D. G. Matthews, and W. M. C. Foulkes. Ab initio solution of the many-electron Schrödinger equation with deep neural networks. *Physical Review Research*, 2(3):033429, 2020. doi: 10.1103/PhysRevResearch.2.033429.

- [37] Jan Hermann, Zeno Schätzle, and Frank Noé. Deep-neural-network solution of the electronic Schrödinger equation. *Nature Chemistry*, 12(10):891–897, 2020. doi: 10.1038/s41557-020-0544-y.
- [38] Sam McArdle, Suguru Endo, Alán Aspuru-Guzik, Simon C. Benjamin, and Xiao Yuan. Quantum computational chemistry. *Reviews of Modern Physics*, 92(1):015003, 2020. doi: 10.1103/RevModPhys.92.015003.
- [39] Yudong Cao, Jonathan Romero, Jonathan P. Olson, Matthias Degroote, Peter D. Johnson, Mária Kieferová, Ian D. Kivlichan, Tim Menke, Borja Peropadre, Nicolas P. D. Sawaya, Sukin Sim, Libor Veis, and Alán Aspuru-Guzik. Quantum chemistry in the age of quantum computing. *Chemical Reviews*, 119(19):10856–10915, 2019. doi: 10.1021/acs.chemrev.8b00803.
- [40] H. Chau Nguyen, Riccardo Zecchina, and Johannes Berg. Inverse statistical problems: from the inverse Ising problem to data science. *Advances in Physics*, 66(3):197–261, 2017. doi: 10.1080/00018732.2017.1341604.
- [41] Nathan Wiebe, Christopher Granade, Christopher Ferrie, and D. G. Cory. Hamiltonian learning and certification using quantum resources. *Physical Review Letters*, 112(19):190501, 2014. doi: 10.1103/PhysRevLett.112.190501.
- [42] Anurag Anshu, Srinivasan Arunachalam, Tomotaka Kuwahara, and Mehdi Soleimanifar. Sample-efficient learning of interacting quantum systems. *Nature Physics*, 17:931–935, 2021. doi: 10.1038/s41567-021-01232-0.
- [43] Timothy J. Lee and Peter R. Taylor. A diagnostic for determining the quality of single-reference electron correlation methods. *International Journal of Quantum Chemistry*, 36(S23):199–207, 1989. doi: 10.1002/qua.560360824.
- [44] Wanyi Jiang, Nathan J. DeYonker, and Angela K. Wilson. Multireference character for 3d transition-metal-containing molecules. *Journal of Chemical Theory and Computation*, 8(2):460–468, 2012. doi: 10.1021/ct2006852.
- [45] Elvira R. Sayfutyarova, Qiming Sun, Garnet Kin-Lic Chan, and Gerald Knizia. Automated construction of molecular active spaces from atomic valence orbitals. *Journal of Chemical Theory and Computation*, 13(9):4063–4078, 2017. doi: 10.1021/acs.jctc.7b00128.
- [46] WooSeok Jeong, Samuel J. Stoneburner, Daniel King, Ruye Li, Andrew Walker, Roland Lindh, and Laura Gagliardi. Automation of active space selection for multireference methods via machine learning on chemical bond dissociation. *Journal of Chemical Theory and Computation*, 16(4):2389–2399, 2020. doi: 10.1021/acs.jctc.9b01297.
- [47] J. P. Coe. Machine learning configuration interaction. *Journal of Chemical Theory and Computation*, 14(11):5739–5749, 2018. doi: 10.1021/acs.jctc.8b00849.
- [48] Chenru Duan, Fang Liu, Aditya Nandy, and Heather J. Kulik. Data-driven approaches can overcome the cost–accuracy trade-off in multireference diagnostics. *Journal of Chemical Theory and Computation*, 16(7):4373–4387, 2020. doi: 10.1021/acs.jctc.0c00358.
- [49] Fang Liu, Chenru Duan, and Heather J. Kulik. Rapid detection of strong correlation with machine learning for transition-metal complex high-throughput screening. *The Journal of Physical Chemistry Letters*, 11(19):8067–8076, 2020. doi: 10.1021/acs.jpcclett.0c02288.

- [50] Jon Paul Janet and Heather J. Kulik. Predicting electronic structure properties of transition metal complexes with neural networks. *Chemical Science*, 8(7):5137–5152, 2017. doi: 10.1039/C7SC01247K.
- [51] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. doi: 10.1038/s42256-024-00832-8.
- [52] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0.
- [53] Adrian Mirza, Nawaf Alampara, Sreekanth Kunchapu, Martiño Ríos-García, Benedict Emoekabu, Aswanth Krishnan, Tanya Gupta, Mara Schilling-Wilhelmi, Macjonathan Okereke, Anagha Aneesh, et al. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nature Chemistry*, 17(7):1027–1034, 2025. doi: 10.1038/s41557-025-01815-x.
- [54] Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models, 2024. URL <https://arxiv.org/abs/2410.05229>.